

Ex. No.: 7

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IPC USING SHARED MEMORY

Aim:

To write a C program to do Inter Process Communication (IPC) using shared memory between sender process and receiver process.

Algorithm:

sender

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Write a string to the shared memory segment using sprintf
5. Set delay using sleep
6. Detach shared memory segment using shmdt

receiver

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Print the shared memory contents sent by the sender process.
5. Detach shared memory segment using shmdt

Program Code:

sender.c

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>

int main() {
    // Create a unique key
    key_t key = ftok("shmfile", 65);

    // Create shared memory segment of size 1024 bytes
    int shmid = shmget(key, 1024, 0666|IPC_CREAT);

    // Attach to the shared memory
```

```

char *str = (char*) shmat(shmid, (void*)0, 0);

// Write data
printf("Enter data to write to shared memory: ");
fgets(str, 1024, stdin);

printf("Data written to shared memory: %s\n", str);

// Detach from shared memory
shmdt(str);
return 0;
}

```

receiver.c

```

#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>

int main() {
    // Use the same key as sender
    key_t key = ftok("shmfile", 65);

    // Locate the shared memory segment
    int shmid = shmget(key, 1024, 0666);

    // Attach to the shared memory
    char *str = (char*) shmat(shmid, (void*)0, 0);

    // Read and print data
    printf("Data read from shared memory: %s\n", str);

    // Detach from shared memory
    shmdt(str);

    // Destroy the shared memory
    shmctl(shmid, IPC_RMID, NULL);

    return 0;
}

```

Sample Output

Terminal 1

```
[root@localhost student]# gcc sender.c -o sender  
[root@localhost student]# ./sender
```

Terminal 2

```
[root@localhost student]# gcc receiver.c -o receiver  
[root@localhost student]# ./receiver  
Message Received: Welcome to Shared Memory  
[root@localhost student]#
```

Result:

Thus, the program is compiled and executed successfully.